DukaPoS deployment & release plan

A practical launch path for the web portal, .NET API, database, and Expo mobile app.

WEB	API + DATA	MOBILE
Vercel Next.js portal	Render Docker API, Postgres, Redis-compatible Key Value	Expo EAS Build and submit store binaries

1 Hosting layout

- **Portal:** deploy the Next.js app from **frontend/** on Vercel. Set `NEXT_PUBLIC_API_URL=https://api.dukapos.co.ke`.
- **API:** deploy the ASP.NET API as a Docker web service on Render.
- **Data:** use paid managed PostgreSQL and Redis-compatible Key Value in the same region as the API. Pick a nearby available region and measure real latency from Kenya.
- **Domains:** point `dukapos.co.ke` to the portal and `api.dukapos.co.ke` to the API; enable HTTPS.

2 Before first deploy

- Fix the Docker build context: Compose currently builds from repository root, but the Dockerfile COPY paths expect projects at the root while the projects live under `src/`.
- Set API CORS to allow the production portal origin. Set production Postgres, Redis, JWT, Daraja, and eTIMS values through the host secret manager.
- Rotate credentials found in committed app settings; replace the default admin password. Keep DB backups and persistent data protection keys.
- Set the public HTTPS M-Pesa callback URL, then validate callbacks and receipts in the provider test environments before live transactions.

3 Mobile distribution

- The app already has Android package ID `co.ke.dukapos.mobile` and iOS bundle ID `co.ke.dukapos.mobile`.
- No `eas.json` is present yet. From `mobile/`, sign in to Expo and run `npx eas-cli build:configure`.
- Android: create a production **.aab**, upload to Google Play Console, start with internal testing, then promote to release.
- iOS: create a production build and send to TestFlight; public release goes through App Store Connect review.
- Set `EXPO_PUBLIC_API_URL=https://api.dukapos.co.ke` in the production build environment.

4 Scanner & receipt printer

- Code/docs indicate web barcode scanning via USB HID and mobile scanning via camera; receipt output uses ESC/POS rendering.
- That does not confirm a specific printer model or connection is tested. Start with a USB HID scanner and verify the printer model/connection (USB, Bluetooth, or network) with a real test receipt before buying in quantity.

Launch domains

Web: dukapos.co.ke
API: api.dukapos.co.ke
Mobile: production EXPO_PUBLIC_API_URL must match the API hostname.

UI redesign: the portal is a Next.js app in `frontend/`. The supplied Stitch project is the design reference. The Stitch MCP is not connected in this Codex session, and its project page did not render here, so no Stitch-based redesign has been applied yet.

Official references

Vercel Next.js: vercel.com/docs/frameworks/full-stack/nextjs | Render regions & free-tier limits: render.com/docs/regions and render.com/docs/free
Expo EAS builds: docs.expo.dev/build/setup/ | Google Play: docs.expo.dev/submit/android/ | TestFlight: docs.expo.dev/submit/testflight/